

Roll No.

TCE-301

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

MECHANICS OF FLUIDS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive the expression for capillary rise in a glass tube. (CO1)

OR

- (b) Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size $0.8 \text{ m} \times 0.8 \text{ m}$ and an inclined plane with angle of inclination 30° . The weight of the square plate is 300 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s . The thickness of oil film is 1.50 mm . (CO1)

2. (a) Differentiate between distorted and undistorted models and explain their merits and limitations. (CO1)

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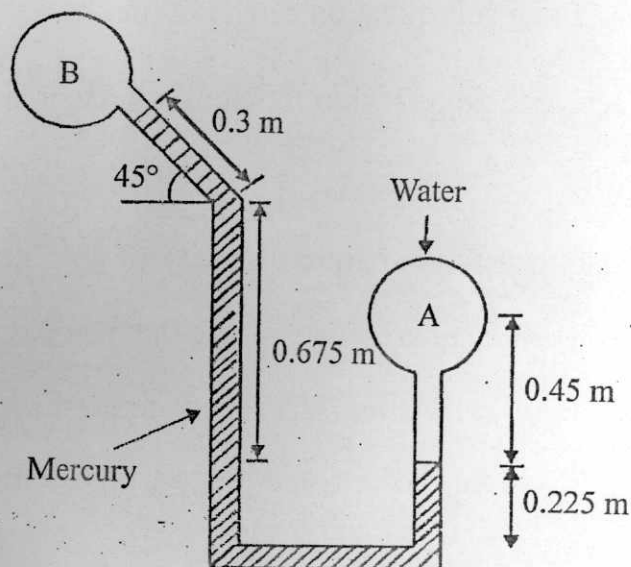
OR

- (b) An oil of specific gravity 0.92 and viscosity 0.3 poise is to be transported at the rate of 2500 litres per second through a 1.20 m diameter pipe tests were conducted on a 12 cm diameter pipe using water at 20°C. If viscosity of water at 20°C is 0.01 poise, then calculate the velocity and rate of flow in model. (CO1)
3. (a) What are various devices adopted for measuring fluid pressure in different conditions ? Illustrate them with the help of a neat diagram. (CO1)

OR

- (b) Determine the pressure difference ($p_A - p_B$) in figure shown below :

(CO2)



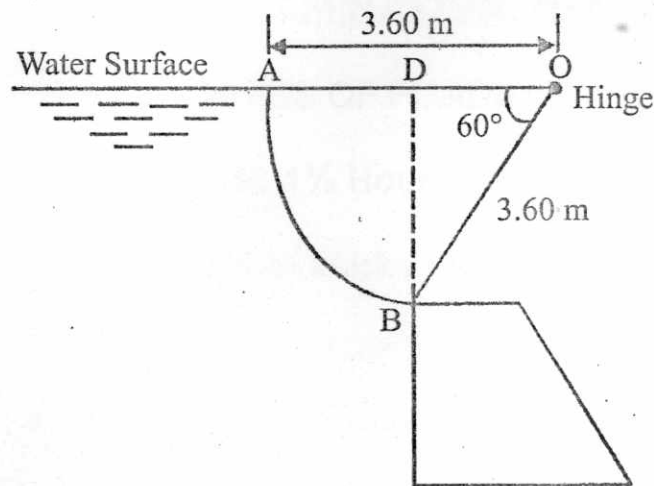
4. (a) Derive the expression for total pressure on a submerged plane surface.

(CO2)

(3)

OR

- (b) A 60° sector gate of 3.60 m radius is mounted on the spillway of a dam as shown in figure. Its hinge and one of its end radial arms are at the same horizontal level as the water surface. What is the magnitude and direction of the resultant pressure on the gate, if the length of the gate is 3.0 m ? (CO2)



5. (a) Describe the various types of flow. (CO2)

OR

- (b) In a three-dimensional incompressible flow the velocity components in the x and y -directions are : $u = x^2 + 2z^2 + 8$ and $v = 2y^2 + z^2 - 6$. Find the velocity component in z -direction. (CO2)